

AC-04: Contraction optimality and a finite-tree obstruction

All tensor powers, arbitrary finite extraction trees,
and an exactly certified upper-bound endpoint

Research continuation prepared with ChatGPT | 14 September 2026

Status: not a solution of AC-04. Neither $\tilde{R}(\text{cw}_2) = 3$ nor its negation is established. The actual tensor bound proved here is $3 \leq \tilde{R}(\text{cw}_2) < 3.923037967879$. The main new results are an all-power optimality theorem for a class of contractions and a rigorous obstruction for a specified family of scalar inequalities. Neither obstruction is a tensor asymptotic-rank lower bound.

Abstract

We audit and extend the supplied AC-04 proof pack. An elementary subspace argument proves that every fully supported weighting of a product of four-term decompositions of the small Coppersmith–Winograd tensor has contraction rank at least 2^n in power n . The bound is attained, and the argument also covers products of four-term border decompositions over a Laurent field. Separately, the scalar value 3.921 is proved feasible for all inequalities generated from the canonical minimum-contraction seeds by arbitrary finite same-orientation extraction trees: leaf powers, arities, and depths need not be equal. A rational invariant proves this without sampling trees. The previous balanced construction gives the slightly sharper actual upper bound above; its infinite scalar threshold is rigorously enclosed near 3.92303796787866460807029148. An exhaustive exact audit additionally classifies all 18 minimum-rank sign contractions in the fixed two-copy construction. These results locate specific limitations of the previous route; they do not establish the exact value requested by AC-04.

1 Question, provenance, and claims

The repository asks, over $\mathbb{C}$, about

$$T = \sum_{i=1}^2 (e_0 \otimes e_i \otimes e_i + e_i \otimes e_0 \otimes e_i + e_i \otimes e_i \otimes e_0). \quad (1)$$

All powers here are grouped three-mode Kronecker powers. The target is

$$\tilde{R}(T) = \lim_{k \rightarrow \infty} R(T^{\otimes k})^{1/k} = 3. \quad (2)$$

AC-04’s current statement explicitly distinguishes this equality from an improved upper bound or a border-rank calculation for one power [1]. The source benchmark is Alman–Li’s $\tilde{R}(T) < 3.931$; their speedup and spectral framework supplies the external machinery used by the inherited construction [2]. The supplied previous pack proved the sharper bound $\tilde{R}(T) < 3.923038$ within that framework [3].

Theorem 1.1 (Actual tensor bound). *For the tensor in (1),*

$$3 \leq \tilde{R}(T) < 3.923037967879.$$

The improvement in Theorem 1.1 is numerically tiny. It is not the main mathematical change in this continuation. Sections 3 and 6 prove the two structural results:

- the contraction rank 2^n cannot be lowered by changing fully supported coefficients, or by changing the four-term factor decompositions before forming their product;
- mixing canonical seed powers and allowing arbitrary finite tree shapes still leaves the formal value 3.921 feasible in the stated scalar relaxation.

The second result does *not* assert $\tilde{R}(T) \geq 3.921$. It does not produce a spectral point with that value, and it does not rule out other tensor constructions.

All mathematical claims below have written derivations. Exact arithmetic checks the finite constants and displayed algebra; it does not replace the all-power or all-tree proofs. The work has not been independently peer reviewed or proof-assistant verified. No research-priority claim is made. The prior pack is retained unchanged as provenance apart from removal of Python caches.

2 Normal form and the lower bound three

Let $P \in (\mathbb{C}^3)^{\otimes 3}$ be the six-permutation tensor

$$P = \sum_{\sigma \in S_3} f_{\sigma(1)} \otimes f_{\sigma(2)} \otimes f_{\sigma(3)}.$$

The invertible map with columns $e_0, e_1 + ie_2, e_1 - ie_2$ satisfies $T = \frac{1}{2}L^{\otimes 3}P$. Consequently T and P have the same rank, border rank, and asymptotic rank. Define

$$V = \begin{pmatrix} 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix} = [v_0 \ v_1 \ v_2 \ v_3].$$

Coefficientwise expansion gives

$$P = \frac{1}{4} \sum_{a=0}^3 v_a^{\otimes 3}. \quad (3)$$

Every flattening of P has rank three: the three matrix slices have disjoint pairs of off-diagonal nonzero entries. Flattening rank multiplies under grouped powers, so $R(P^{\otimes k}) \geq 3^k$. Submultiplicativity of tensor rank proves existence of the limit and gives $3 \leq \tilde{R}(P) \leq 4$.

A fact needed for the new contraction theorem is stronger than conciseness: every nonzero matrix slice of P has rank at least two. Such a slice has form

$$\begin{pmatrix} 0 & c & b \\ c & 0 & a \\ b & a & 0 \end{pmatrix}, \quad (a, b, c) \neq (0, 0, 0). \quad (4)$$

A nonzero entry, for example a , gives a principal minor $-a^2 \neq 0$. In particular $R(P)$ cannot be three: a concise rank-three decomposition in three-dimensional factors has a nonzero rank-one slice. Together with (3), this proves $R(P) = 4$.

The attaining contraction in the previous construction is

$$V \operatorname{diag}(1, 1, -1, -1) V^\top = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 4 \\ 0 & 4 & 0 \end{pmatrix}, \quad (5)$$

which has rank two. Tensoring this identity gives contraction rank 2^n with all 4^n coefficients nonzero.

3 An all-power contraction optimality theorem

This section gives an analytic result beyond the earlier finite sign search. The argument is a rank bound for a diagonal matrix plus a perturbation whose image lies in a specified proper subspace.

Lemma 3.1 (Product-matrix contraction). *For $i = 1, \dots, n$, let $A_i, B_i \in \mathbb{F}^{3 \times 4}$ be matrices whose first three columns form bases, and whose fourth columns are nonzero. Let $w \in (\mathbb{F} \setminus \{0\})^{4^n}$. Then*

$$\operatorname{rank}_{\mathbb{F}} \left[(A_1 \otimes \cdots \otimes A_n) \operatorname{diag}(w) (B_1 \otimes \cdots \otimes B_n)^\top \right] \geq 2^n. \quad (6)$$

Here $\mathbb{F}$ is any field, and the bases may be changed independently on the left and right.

Proof. Multiply each A_i on the left by the inverse of its first three columns, and do the same for B_i . These operations preserve the rank of the displayed matrix. We may therefore write

$$A_i = [I_3 \mid a_i], \quad B_i = [I_3 \mid b_i], \quad a_i, b_i \neq 0.$$

Index the 4^n columns by words in $\{0, 1, 2, 3\}^n$. The columns indexed by $\{0, 1, 2\}^n$ are the standard basis on both sides. Their contribution is a nonsingular diagonal matrix D of size 3^n , because all corresponding entries of w are nonzero.

Every remaining left column belongs to

$$W = \sum_{i=1}^n (\mathbb{F}^3)^{\otimes(i-1)} \otimes \operatorname{span}(a_i) \otimes (\mathbb{F}^3)^{\otimes(n-i)}.$$

Choose a two-dimensional complement E_i to $\operatorname{span}(a_i)$ in each factor. The tensor-product direct-sum expansion shows that W is the sum of all blocks except $E_1 \otimes \cdots \otimes E_n$. Thus

$$\dim W = 3^n - 2^n.$$

Write the sum of the remaining rank-one terms as Q . Its image lies in W , so $\operatorname{rank} Q \leq 3^n - 2^n$. Since D is nonsingular,

$$\operatorname{rank}(D + Q) \geq \operatorname{rank} D - \operatorname{rank} Q \geq 3^n - (3^n - 2^n) = 2^n.$$

The right vectors in the exceptional terms need no further structure for this image-space argument. This proves (6). $\square$

Lemma 3.2 (Full-spark factor matrices). *In every exact four-term decomposition of P , each factor matrix has every three columns linearly independent. The same assertion holds over a Laurent field for a four-term border decomposition $P + O(\lambda)$.*

Proof. In the exact case, suppose three columns in the first factor are dependent. Conciseness forces the four columns together to span a three-dimensional space. A nonzero functional annihilating the dependent three therefore leaves only the fourth term in the contraction. This gives a nonzero rank-one slice of P , contradicting (4). The other two modes are identical.

For the border case, suppose three columns are dependent over $\mathbb{C}((\lambda))$. Choose a nonzero Laurent-series functional annihilating them, and multiply it by a power of λ so that it equals $\ell_0 + O(\lambda)$ with $\ell_0 \neq 0$. Its contraction with the represented tensor has rank at most one over the Laurent field, so every 2×2 minor vanishes identically. Taking the constant term in

$$(\ell_0 + O(\lambda)) \lrcorner (P + O(\lambda)) = \ell_0 \lrcorner P + O(\lambda)$$

shows $\text{rank}(\ell_0 \lrcorner P) \leq 1$. This is again impossible by (4). Puiseux parameters cause no change after reparameterization. $\square$

Theorem 3.3 (Optimal contraction rank in every product power). *Take, in each of n factors, any four-term exact decomposition of P , or any four-term border decomposition of P over a common Laurent field. Form their product decomposition with 4^n terms. For every choice of nonzero weights on all these terms, the contraction in the two remaining modes has rank at least 2^n over its coefficient field. This bound is attained by the product of (3) with weights from (5).*

Proof. Lemma 3.2 verifies the hypotheses of Lemma 3.1 for any two factor modes. The latter gives the lower bound. Equation (5) and multiplicativity of matrix rank give attainment. $\square$

For the particular V , the reduction used in the proof is explicit:

$$\begin{pmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 0 & -1/2 \\ 0 & 1/2 & -1/2 \end{pmatrix} V = \begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \end{pmatrix}.$$

The accompanying code checks this identity and the exceptional-space dimensions for $n = 1, 2, 3$. The analytic direct-sum argument, not those finite checks, proves the statement for every n .

Scope. The theorem rules out improving the contraction parameter $s = 2^n$ by changing fully supported coefficients in *product four-term factor decompositions*. It does not cover a genuinely non-product decomposition of $P^{\otimes n}$, a different extraction operation, or additional use of core tensor structure. It is not a lower bound on $R(P^{\otimes n})$ or $\tilde{R}(P)$.

4 Extraction seeds and the scalar relaxation

Write D_r for a diagonal tensor of rank r , and L_s for an identity slice of dimensions $(s, s, 1)$. Degeneration is denoted $K \leq S$, meaning that S degenerates to K . We use the full, isolated one-slice extraction mechanism from the earlier pack [3], an instance of Alman–Li’s construction [2, Sections 5–6].

The linear-algebraic mechanism is as follows. Suppose a third-mode source functional contracts the source to a nonsingular matrix H of size M , while the first two target maps A, B have d_A, d_B rows and satisfy $AHB^\top = 0$. Restrict the rows on the first side to $\ker(HB^\top)$ and on the second to the corresponding left annihilator of AH . The compressed pairing has rank at least $M - d_A - d_B$. Augmenting A, B by rows realizing that rank yields an auxiliary identity slice. A blockwise

degeneration keeps the core and auxiliary slice and removes the cross terms. The new third-mode coordinate is isolated: all cross blocks using it vanish identically. Its source contraction still has rank M , which is the fullness condition. These statements also hold over a rational-function field, allowing composition and iteration. The previous report supplies the explicit block weights and the complete kernel proof [3].

For the n th power, (3) and (5) give

$$r_n = 4^n, \quad s_n = 2^n, \quad d_n = 3^n, \quad M_n = 4^n + 2^n, \quad t_n = M_n - 2d_n. \quad (7)$$

Adding L_{s_n} cancels the rank- s_n contracted image. The source contraction is nonsingular of size M_n , so for $n \geq 2$ one obtains

$$P^{\otimes n} \oplus L_{t_n} \leq D_{r_n} \oplus L_{s_n}. \quad (8)$$

The slice is full and isolated. At $n = 3$ the exact inherited matrix audit reproduces a nonsingular source matrix of size 72 and an auxiliary slice of rank 18, including an 18×18 minor of determinant -1 .

We use the standard spectral-duality facts in [2, Section 4]: a spectral point ϕ is additive, multiplicative, degeneration-monotone, and satisfies $\phi(D_r) = r$; moreover

$$\tilde{R}(P) = \max_{\phi} \phi(P).$$

Its three oriented identity-slice exponents obey

$$\phi(L_s^{(i)}) = s^{\theta_i}, \quad 0 \leq \theta_i \leq 1, \quad \theta_1 + \theta_2 + \theta_3 \geq 2. \quad (9)$$

These are imported structural theorems, not code-verified claims. Symmetry of P supplies three oriented copies of the construction. At least one exponent is at least $2/3$.

For the obstruction theorem we make the *formal* assignment

$$z = \frac{3921}{1000}, \quad \theta_1 = \theta_2 = \theta_3 = \vartheta := \frac{2}{3}. \quad (10)$$

We prove it satisfies a specified infinite collection of necessary scalar inequalities. We do not assert that it extends to a genuine ϕ .

5 Arbitrary finite extraction trees

Definition 5.1 (The tree family). A leaf is any seed (8) with integer $n \geq 2$. A vertex takes any finite number $m \geq 2$ of previously constructed children, all with the same slice orientation. Tensor the children, remove the tensor product of their distinguished slices, and re-extract a full isolated slice of the canonical size $M - 2d$. No additional restriction or compression of the remaining core is performed. Arity, depth, and leaf powers may differ. The three cyclic copies of this family are all included as separate scalar constraints.

At a child write $U_i = K_i \oplus L_{t_i} \leq S_i$, where K_i has first two dimensions d_i , the source has those dimensions $M_i = t_i + 2d_i$, and its formal scalar cost is h_i . Let p_i denote the formal value of U_i .

Proposition 5.2 (Tree recurrences). *At a vertex with m children, the parameters are*

$$\begin{aligned} M &= \prod_i M_i, & d &= \prod_i (d_i + t_i) - \prod_i t_i, & t &= M - 2d, \\ h &= \prod_i h_i, & p &= \prod_i p_i - \left(\prod_i t_i \right)^\vartheta + t^\vartheta. \end{aligned} \quad (11)$$

These are valid necessary spectral constraints $p \leq h$ for the corresponding full isolated degenerations.

Proof. The first two target dimensions before removal are $\prod_i (d_i + t_i)$. Removing the pure slice block deletes $\prod_i t_i$ coordinates in each mode. The remaining core therefore has first two dimensions d in (11). The product of the distinguished source functionals has rank M and is isolated. After removing the slice coordinates its contracted target is zero, so the extraction mechanism gives $t = M - 2d$. Fullness and isolation are preserved for the next vertex, just as in the binary iteration of the prior report. The scalar formulas follow by multiplying the child values and subtracting only the removed summand before adding the replacement. In particular this subtraction represents an actual direct-sum decomposition; it is not a general subtraction rule for rank. $\square$

Normalize at every vertex by

$$a = t/M, \quad H = h/M^\vartheta, \quad u = p/M^\vartheta. \quad (12)$$

Then the recurrence becomes

$$\begin{aligned} a' &= 1 - 2 \prod_i \frac{1 + a_i}{2} + 2 \prod_i a_i, \\ H' &= \prod_i H_i, \quad u' = \prod_i u_i + \Delta_m, \quad \Delta_m = (a')^\vartheta - \left(\prod_i a_i \right)^\vartheta. \end{aligned} \quad (13)$$

For $a_i \in [0, 1]$, one has $0 \leq \Delta_m \leq 1$. Indeed, $\prod_i a_i \leq a' \leq 1$. The first inequality follows from $2 \prod_i (1 + a_i)/2 \leq 1 + \prod_i a_i$, a multi-affine inequality verified at vertices $a_i \in \{0, 1\}$. The second follows from $\prod_i (1 + a_i)/2 \geq \prod_i a_i$.

6 A rigorous obstruction at the formal value 3.921

Theorem 6.1 (All finite trees remain feasible). *For the formal assignment (10), every tree in Definition 5.1 satisfies $p < h$. Consequently this entire family of scalar inequalities, together with (9) and $3 \leq z \leq 4$, cannot imply $z < 3.921$, and in particular cannot imply $z = 3$.*

We prove a uniform invariant. Set

$$C = \frac{49}{1000}, \quad C_2 = \frac{1}{5}. \quad (14)$$

At every node we claim

$$\frac{1}{10} \leq a \leq 1, \quad H > \frac{5}{2}, \quad 0 \leq u \leq H - C. \quad (15)$$

A leaf of power two has the stronger deficit $u \leq H - C_2$. A leaf of power three has $H > 3929/1000$ and $a = 1/4$. Every other node, except a single power-two leaf, has $H > 6$ unless it is precisely a single power-three leaf.

6.1 Every leaf, including all sufficiently large powers

For a leaf the deficit assertion is equivalent to

$$z^n + t_n^{2/3} + C_n M_n^{2/3} < 4^n + (2^n)^{2/3}, \quad C_n = \begin{cases} 1/5, & n = 2, \\ 49/1000, & n \geq 3. \end{cases} \quad (16)$$

Exact rational cube-root intervals verify (16) for $n = 2, 3, 4, 5$. The following rounded positive slacks are provided only for orientation; the full rational enclosures are in the certificate.

n	t_n	Right minus left in (16)
2	2	0.0845874483...
3	18	0.0012728363...
4	110	0.9675770490...
5	570	33.4555806075...

For $n \geq 6$, use $t_n \leq M_n \leq (9/8)4^n$ and $C < 1/20$. The rational inequality

$$(21/20)^3 (9/8)^2 < (5/4)^3$$

implies $(1 + C)M_n^{2/3} < (5/4)4^{2n/3}$. Also the exact rational check

$$1 - (z/4)^6 > 5/64$$

and monotonicity in n give

$$1 - (z/4)^n > \frac{5}{4} 4^{-n/3} \quad (n \geq 6).$$

Thus $z^n + (1 + C)M_n^{2/3} < 4^n$, which is stronger than (16). This proves the leaf inequality for *every* $n \geq 2$.

For the other invariant bounds, $a_n = 1 - 2 \cdot 3^n / (4^n + 2^n)$ increases from $a_2 = 1/10$; the ratio subtracted decreases because $2^n < 4^n$. At $n = 2$, the inequalities $16^{1/3} > 5/2$ and $400^{1/3} < 37/5$ imply $H_2 > (37/2)/(37/5) = 5/2$. At $n = 3$,

$$H_3 = 68/72^{2/3} > 3929/1000$$

is checked by cubing. For $n \geq 4$, $M_n \leq (17/16)4^n$ gives

$$H_n^3 \geq \frac{4^n}{(17/16)^2} \geq \frac{65536}{289} > 216,$$

so $H_n > 6$. Positivity of u is immediate at every leaf.

6.2 The normalized slice interval is preserved

The expression for a' in (13) is multi-affine. Its extrema on $[1/10, 1]^m$ occur at vertices. If k coordinates are $1/10$, its value is

$$f(k) = 1 - 2(11/20)^k + 2(1/10)^k.$$

Here $f(0) = 1$, $f(1) = 1/10$, and for $k \geq 1$,

$$f(k+1) - f(k) = \frac{9}{10} ((11/20)^k - 2(1/10)^k) > 0.$$

Thus $a' \geq 1/10$. We already know $a' \leq 1$ and $\Delta_m \geq 0$. Every non-leaf has $H' = \prod_i H_i > (5/2)^2 > 6$, and $u' \geq 0$.

6.3 Binary vertices: four exhaustive cases

For two children,

$$\Delta(a, b) = \left(ab + \frac{(1-a)(1-b)}{2} \right)^{2/3} - (ab)^{2/3}. \quad (17)$$

This is nonincreasing in each variable on $[1/10, 1]$. To see this, put $S = ab + (1-a)(1-b)/2$. Differentiation in a gives

$$\frac{\partial \Delta}{\partial a} = \frac{2}{3} \left[S^{-1/3} \frac{3b-1}{2} - b(ab)^{-1/3} \right] \leq 0.$$

If $b \leq 1/3$, the sign is immediate. Otherwise $(3b-1)/2 \leq b$ and $S \geq ab$, which gives the same sign. Symmetry handles b . Exact interval arithmetic establishes

$$\Delta(1/4, 1/4) < 1/3, \quad \Delta(1/4, 1/10) < 17/40, \quad \Delta(1/10, 1/10) < 51/100. \quad (18)$$

If the two child deficits are c_1, c_2 , then

$$H_1 H_2 - u_1 u_2 \geq c_1 H_2 + c_2 H_1 - c_1 c_2.$$

To retain deficit C after adding Δ , it suffices that the right side exceed $C + \Delta$. The following four rational comparisons cover all binary vertices:

$$\begin{aligned} C(2 \cdot 3929/1000 - 1) - C^2 &> 1/3 && \text{(two power-three leaves),} \\ C(39/10 + 6 - 1) - C^2 &> 17/40 && \text{(one such leaf; no power-two leaf),} \\ C(12 - 1) - C^2 &> 51/100 && \text{(neither exceptional leaf),} \\ C_2(5/2) + C(5/2) - C_2 C - C &> 51/100 && \text{(at least one power-two leaf).} \end{aligned}$$

The second case uses $a = 1/4$ for its power-three child; all other relevant a values are at least $1/10$. In the last case we use the stronger child deficit C_2 . The certificate checks each strictly positive rational margin.

6.4 All larger arities

We only need $\Delta_m \leq 1$. The product gap $\prod_i H_i - \prod_i (H_i - c_i)$ is increasing in each H_i when $H_i > c_i > 0$. For arity three, with a power-two leaf, it is greater than

$$(5/2)^3 - (5/2 - C_2)(5/2 - C)^2 > 1 + C.$$

Without a power-two leaf, every child has $H_i > 39/10$, and the gap is greater than

$$(39/10)^3 - (39/10 - C)^3 > 1 + C.$$

For $m \geq 4$, using only $H_i > 5/2$ and $c_i \geq C$, the gap is at least

$$mC(5/2 - C)^{m-1}.$$

This follows by expanding the product with $H_i = (H_i - C) + C$ and keeping its m terms linear in C . At $m = 4$ the displayed expression is greater than $1 + C$, and it increases thereafter. These rational comparisons are also included in the exact certificate.

All vertices therefore preserve (15). Induction over every finite tree proves Theorem 6.1, because $u < H$ implies $p < h$. The three oriented families are simultaneously satisfied by (10). Any consequence obtained solely by valid algebraic combination of these scalar inequalities remains satisfied at this assignment.

6.5 What the obstruction does and does not cover

The theorem is an all-depth, all-arity statement, not a finite numerical search. It also allows arbitrarily large and mixed initial powers. Its seed parameters are exactly (7), with the optimal product-contraction rank $s_n = 2^n$ supplied by Theorem 3.3.

It does *not* characterize the entire tensor asymptotic spectrum. It does not include fresh extraction after mixing different slice orientations, additional core compression, simultaneous multiple slices, non-product power decompositions, or all possible larger source paddings. Ordinary sums or products of already established scalar inequalities add no contradiction, but a new tensor identity could. The value 3.921 is a feasible point of a relaxation, not a counterexample to AC-04.

7 The sharpened upper bound and the balanced limit

We now complete Theorem 1.1 and distinguish the numerical threshold of the old balanced recurrence from the actual tensor asymptotic rank. Start at power three and repeatedly take two identical children:

$$(d_0, t_0, M_0) = (27, 18, 72), \quad (19)$$

$$d_{j+1} = d_j^2 + 2d_j t_j, \quad t_{j+1} = t_j^2 + 2d_j^2, \quad M_{j+1} = M_j^2.$$

In particular

$$d_j = \frac{72^{2^j} - (-9)^{2^j}}{3}, \quad t_j = \frac{72^{2^j} + 2(-9)^{2^j}}{3}. \quad (20)$$

For $z \in [0, 4]$ and $\theta > 0$, put

$$h(\theta) = 64 + 8^\theta, \quad p_0(z, \theta) = z^3 + 18^\theta, \quad (21)$$

$$p_{j+1}(z, \theta) = p_j(z, \theta)^2 - t_j^{2\theta} + t_{j+1}^\theta, \quad q_j = p_j / h(\theta)^{2^j}.$$

Every spectral value $z = \phi(P)$ satisfies $q_j(z, \theta) \leq 1$ in the chosen orientation. As shown in the prior report and rechecked here, q_j increases in θ for $z \in [0, 4]$. For completeness, the numerator of $\partial_\theta q_0$ has positive terms arranged as

$$64 \cdot 18^\theta \log 18 - z^3 \cdot 8^\theta \log 8 + 18^\theta \cdot 8^\theta \log(18/8) > 0.$$

Further,

$$q_{j+1} = q_j^2 + E_j(\theta), \quad E_j(\theta) = \frac{t_{j+1}^\theta - t_j^{2\theta}}{h(\theta)^{2^{j+1}}}. \quad (22)$$

Because $t_j > 8^{2^j}$ and $t_{j+1} > t_j^2$, logarithmic differentiation gives $E'_j/E_j > 2 \log t_j - 2^{j+1} \log 8 > 0$. Positivity then proves monotonicity of q_j inductively. One of the three orientations has $\theta \geq 2/3$, so every spectral value obeys

$$q_j(z, 2/3) \leq 1, \quad h(2/3) = 68. \quad (23)$$

No sampled parameter grid is used for this reduction.

7.1 Exact finite-depth certification

At the rational input $b = 3.923037967879$, the checker proves

$$q_4(b, 2/3) > 1.$$

Its value is approximately 1.0000000000036012972797655, but the strict comparison uses rational interval endpoints. Since q_4 is strictly increasing in $z \geq 0$, (23) and spectral duality give $\tilde{R}(P) < b$. Together with Section 2, this proves Theorem 1.1.

The code uses the following exact enclosure operation for a nonnegative rational x . Let $D = 10^{80}$ and

$$k = \left\lfloor \sqrt[3]{\lfloor x^2 D^3 \rfloor} \right\rfloor.$$

Then $k/D \leq x^{2/3} < (k+1)/D$, unless equality at the lower endpoint is recognized exactly. Integer cube-root comparisons and rational interval arithmetic propagate these enclosures through the recurrence. Full rational endpoints are stored in `results/upper_certificate.json`; decimal displays there are informational and are not the decision procedure.

7.2 A certified limit for this scalar family

Let β_j be the unique nonnegative root of $q_j(z, 2/3) = 1$. The polynomial p_j has nonnegative coefficients in z^3 and positive leading coefficient. Its constant term is $t_j^{2/3}$, so $q_j(0, 2/3) < 1$ because $72^{2/3} < 68$. Thus the root exists and is unique. At $z = \beta_j$, $q_{j+1} = 1 + E_j > 1$, whence $\beta_{j+1} < \beta_j$. Therefore $\beta_\infty = \inf_j \beta_j$ exists.

Theorem 7.1 (Balanced scalar threshold). *The threshold of all the balanced power-three constraints satisfies*

$$\begin{aligned} 3.92303796787866460807029147 &< \beta_\infty \\ &< 3.92303796787866460807029149. \end{aligned}$$

This is not an identification of $\tilde{R}(P)$ with β_∞ .

Proof. Write the two displayed rationals as b_- and b_+ . Exact arithmetic verifies $q_5(b_+, 2/3) > 1$, hence $\beta_\infty \leq \beta_5 < b_+$. For the lower bound set $c = 1 - 10^{-26}$. The checker proves

$$q_5(b_-, 2/3) < c < 1, \quad (9/34)^{64} < c(1 - c).$$

For all $j \geq 5$, using $t_{j+1} < M_{j+1}$ and $72^{2/3} < 18$ gives

$$0 < E_j(2/3) < \left(\frac{72^{2/3}}{68} \right)^{2^{j+1}} < \left(\frac{9}{34} \right)^{2^{j+1}} \leq (9/34)^{64} < c(1 - c).$$

Thus $q_{j+1} < c^2 + c(1 - c) = c$ whenever $q_j < c$. This proves feasibility at every $j \geq 5$. Earlier q_j are also below one, since $q_{j+1} = q_j^2 + E_j$ with $E_j > 0$. The strict margin at $j = 5$ persists for some slightly larger input by continuity, and the same tail induction applies to that input. Consequently $\beta_\infty > b_-$. $\square$

The finite fifth-step endpoint is also enclosed exactly:

$$3.92303796787866460807029147905217484245 < \beta_5 \\ < 3.92303796787866460807029147905217484247.$$

More balanced squarings cannot turn this family into a proof of value three; Theorem 7.1 is a rigorous statement about that failure, rather than an inference from apparent numerical convergence. The stronger Theorem 6.1 covers many other tree choices as well, though with the lower obstruction value 3.921.

8 Exhaustive two-copy sign classification

The following finite result supplements the all-power analytic theorem. Let $A = V \otimes V$ and, for $w \in \{+1, -1\}^{16}$, set

$$M_w = A \operatorname{diag}(w) A^\top.$$

Theorem 8.1 (The 18 sign minimizers). *Among the 65,536 sign vectors, the minimum complex rank of M_w is four. Exactly 18 vectors attain it. They are precisely $w = a \otimes b$, where $a, b \in \{+1, -1\}^4$ each have two positive and two negative coordinates.*

Proof. The minimum lower bound follows already from Theorem 3.3. For the classification, the supplied exhaustive checker forms each integer matrix and computes its rank modulo 101. It finds precisely 18 matrices of modular rank four, and checks that their masks equal the proposed set. All other matrices have modular rank at least five, hence complex rank at least five. Each proposed factor has exact contraction rank two: up to sign it is one of the three nonconstant characters of $\mathbb{F}_2^2$. Therefore each proposed product has exact complex rank four. The number of distinct products is $6 \cdot 6/2 = 18$, since simultaneous negation of the factors does not change their product. This proves the classification. $\square$

The complete *modular* rank histogram is

Rank 4	Rank 5	Rank 6	Rank 7	Rank 8	Rank 9
18	288	2,964	13,128	26,496	22,642

It is not claimed that this histogram equals the complex-rank histogram beyond the minimum-rank classification. Reduction modulo a prime can lower rank, never increase it for an integer matrix.

The standard-library Python checker uses a Walsh transform to form the matrices, with direct-matrix regression tests verifying the ordering. An independent C++ implementation forms the matrices directly and reproduces both the histogram and the 18 masks. Its source and output are supplied; no executable is needed in the package. The checker is a complete finite computation, not a random search. The all-power result does not depend on this enumeration.

9 Remaining gap and routes actually tested

Neither the upper bound nor either obstruction proves (2). A positive resolution still needs, for example,

$$\forall \varepsilon > 0 \exists k \geq 1 : \quad R(T^{\otimes k}) < (3 + \varepsilon)^k. \quad (24)$$

The equivalence follows from submultiplicativity and the flattening lower bound. A sequence of suitable border-rank bounds could also suffice through the standard asymptotic-rank/border-rank comparison. A negative resolution would need a genuine asymptotic lower bound greater than three, not a feasible point of the restricted scalar relaxation.

Exploratory computations compared initial powers 2 through 12, mixed seed pairs, unequal binary trees for selected leaf counts, and direct arities 2 through 12. They did not yield a better construction than the balanced power-three seed at the displayed precision. These finite computations are *not* used to prove global optimality. The all-tree invariant replaces that possible overinterpretation with a precisely delimited theorem.

The all-power contraction theorem eliminates a different tempting escape: allowing arbitrary complex weights, instead of only signs, or choosing another four-term decomposition of each factor cannot lower the product contraction rank. It remains possible to leave that product-decomposition class, use multiple or differently oriented slices together, or exploit additional core restrictions. None of those stronger constructions is supplied here.

Two relevant external results must not be overread. Conner–Huang–Landsberg’s border-rank result for $T^{\otimes 2}$ gives 16, but a fixed-power equality alone does not determine (2) [4]. The polynomial characterization of asymptotic rank gives Zariski-closed sublevel sets and well-ordering results [5]; it does not turn this sequence of *upper bounds* into actual attainable rank values or provide a numerical gap forcing a bound near 3.923 to equal 3. No such inference is used.

The known implication $\tilde{R}(T) = 3 \Rightarrow \omega = 2$ explains the significance of the target [2, Theorem 1.2]; it is not a reason to reject an attempt at it. The specific reason this continuation does not close the problem is that no new construction or spectral argument establishes (24) or its negation. The repository’s exact-value item should not be represented as solved by this package.

10 Reproduction and audit record

Run the essential verification with Python’s standard library:

```
python code/run_all.py
```

It regenerates four certificates, checks all 65,536 sign patterns, runs 18 new regression tests, and reruns the eight inherited tests. To rerun the previous exact SymPy matrix audit as well, install the optional dependency and run

```
python -m pip install sympy
python code/run_all.py --with-algebra
```

The recorded run included the optional matrix audit; all checks passed. Assertions must remain enabled: do not pass Python’s `-O` option.

File	Purpose
<code>code/exact.py</code>	Integer cube roots and rational intervals.
<code>code/certify_upper.py</code>	Actual upper bound and balanced-limit enclosure.
<code>code/certify_tree_barrier.py</code>	Finite constants supporting the all-tree induction.
<code>code/contraction_lower_bound.py</code>	Finite algebra checks for the analytic all-power theorem.
<code>code/sign_audit.py</code>	Complete finite sign classification.
<code>code/test_round2.py</code>	18 new regression tests.
<code>code/optional/</code>	Independent C++ sign audit and numerical explorations.
<code>results/</code>	Exact certificates, verification logs, and labeled exploratory output.
<code>prior_round/</code>	The supplied earlier proof pack for provenance.

The reported passing computations establish their stated finite checks only. The all-power and arbitrary-tree conclusions additionally require the mathematical proofs in this document. There is no claim of independent review, machine-checked formal proof, a complete literature priority search, or a full AC-04 solution. The package does not alter the repository.

References

- [1] OpenProblemsInNLA, *AC-04: Minimal asymptotic rank of the small Coppersmith–Winograd tensor*. Repository statement last checked 10 September 2026; accessed 14 September 2026. [Canonical problem statement](#).
- [2] J. Alman and B. Li, *Asymptotic Rank Speedup Theorems, Revisited*. CCC 2026, LIPIcs 383, 36:1–36:42, 23 July 2026. [doi:10.4230/LIPIcs.CCC.2026.36](#). Also [arXiv:2605.21738](#), 20 May 2026. Sections 4–7 supply the spectral and extraction framework.
- [3] *AC-04: A certified upper-bound refinement*. Previous research pack supplied in this conversation, request date 13 September 2026. Preserved in `prior_round/report.pdf` and its editable source. Research work product, not independently reviewed literature.
- [4] A. Conner, H. Huang, and J. M. Landsberg, *Bad and good news for Strassen’s laser method: Border rank of the 3×3 permanent and strict submultiplicativity*. [arXiv:2009.11391](#).
- [5] M. Christandl, K. Hoeberechts, H. Nieuwboer, P. Vrana, and J. Zuiddam, *Asymptotic tensor rank is characterized by polynomials*. [arXiv:2411.15789v2](#), 7 June 2026.